

NORA Latency & Call-Flow Plan

As of 2026-09-17 | Backend fine-tuning plan (no code changed yet)

NORA can likely get under 1 second from the caller finishing to NORA's first audio, and its call flow should move to a code-driven policy layer with a much smaller prompt. No code has been changed yet; this is the plan.

Summary

- **Latency:** measure every stage first, then apply safe wins (connection warm-up, shorter prompt, fixed lines for fixed situations), then flagged changes (always-on speculative LLM, better end-of-turn detection, Rime WebSocket streaming).
- **Edge cases:** a priority ladder from compliance stops (P0) to normal flow (P5), plus a catalog of about 60 situations, from voicemail to hold requests to spam prevention.
- **Flow:** one policy engine makes every decision; the LLM only phrases replies inside the current stage. This removes the triple-sourced follow-ups, the three copies of email confirmation, and the duplicated prompt blocks.
- **Current use case:** payment and service reminders (the uncommitted pivot from insurance benefits verification).

Part 1: Latency

The biggest likely costs are end-of-turn detection, the serial intent call, and a large system prompt. The figures below are estimates from reading the code; Step 0 replaces them with measured numbers.

Where the time likely goes

Stage	Current behaviour (code ref)	Estimated cost
End-of-turn detection	Deepgram endpointing=600, then finals plus 0.2 s of quiet, plus <code>audio_chunk_count > 10</code> (<code>stt.py:64</code> , <code>main.py:971</code>)	700-900 ms
Intent classification	Serial Groq 8B call when regex shortcuts miss (<code>main.py:1260</code>)	0-400 ms
LLM first token	70B model, ~2.5-3k-token prompt rebuilt every turn, duplicated voice block (<code>language.py:1257</code> , <code>:1298</code>)	250-600 ms
First speakable piece	Waits for punctuation plus a minimum word count (<code>llm.py:388</code>)	100-300 ms
TTS first byte	New HTTP POST per piece; prefetch downloads the whole clip (<code>tts.py:514</code>)	200-500 ms

Step 0: Measure first (no behaviour change)

Log six timestamps per turn in `log_turn`: end of user speech, transcript ready, intent done, LLM first token, first TTS byte, first audio frame sent. Collect p50 and p95 across about 30 test calls, and judge every later change against them.

Step 1: Safe wins (same behaviour)

- 1 **Warm the Rime connection** at session start, like the existing Groq warm-up (`main.py:638`).
- 2 **Shorten the prompt and put fixed content first.** Persona, voice and compliance rules stay identical each turn; stage, intent and limits go last. This speeds up the first token and allows provider-side prompt caching where Groq supports it (not confirmed for this model).
- 3 **Feed Deepgram during THINKING.** Audio only reaches Deepgram in LISTENING (`main.py:861`), so words spoken while NORA thinks, or in the 0.5 s playback padding, are lost and cause re-asks.
- 4 **Replay the last reply for "say that again"** without calling the LLM.
- 5 **Use fixed, pre-recorded lines** for do-not-call, wrong number, goodbye and hold. Today exits still call the LLM and then sleep 1.5-2 s (`main.py:1470`, `:1492`).

6 **Replace fixed goodbye sleeps** with the real end of playback.

Step 2: Moderate changes (behind feature flags)

- 1 **Start the LLM on every turn** while the classifier runs in parallel. Restart only when the classifier returns an exit, refusal, escalation or do-not-call intent.
- 2 **Improve end-of-turn detection.** Read Deepgram's `speech_final` flag (only `is_final` is used now, which can fire mid-sentence). Try nova-3 with endpointing at 300-400 ms, keeping the existing incomplete-phrase filters as a guard.
- 3 **Move Rime to WebSocket streaming:** one socket per call, text sent as the LLM generates it, with word timestamps that show what was spoken before a barge-in.
- 4 **Flush the first phrase sooner:** at the first comma once it has 3 or more words.
- 5 **Use real playback-finished signals** instead of `bytes/32000 + 0.5 s`: a `PLAYBACK_DONE` message from the browser and Twilio mark events on phone calls.

Step 3: Structural changes

- 1 **Route turns by type:** a fast small model for simple and templated turns, the 70B only for explanations and objections. Benchmark quality before switching.
- 2 **Improve the phone audio path.** Replace `np.repeat` upsampling (`twilio_bridge.py:218`) with proper resampling, or send 8 kHz mu-law straight to Deepgram. Run the bridge in-process instead of looping back through `ws://127.0.0.1/ws/voice`.
- 3 **Host near your providers.** Put the backend in the same region as Deepgram, Groq and Rime; target under 1 s end to end.

Part 2: Edge cases and escalation

Every turn is handled by the highest-priority rule that matches; compliance stops always win. An AI voice counts as an "artificial voice" under the TCPA, so calls need prior consent, identification and an opt-out, and several states require AI disclosure. The current instruction for NORA to deny being an AI (`language.py:969`) should become honest disclosure. If reminders involve money owed on behalf of another organization, FDCPA and Regulation F also apply: 8am-9pm local time, 7 call attempts per 7 days, limited voicemail content, and no disclosure to third parties. Confirm with counsel which rules apply.

Priority ladder

Priority	Category	Handling
P0 Compliance stop	Do-not-call, revoked consent, attorney, bankruptcy, cease-and-desist, deceased, wrong number	Fixed line, record it, end call. No LLM, no follow-up
P1 Safety	Threats, self-harm, medical emergency, abuse	Fixed safety line, human escalation, end
P2 Human needed	Asks for a human, disputes amount, hardship, payment plan, complaint, legal question	Warm transfer if an agent is free, otherwise queue or callback
P3 Right-party problem	Third party answers, account holder unavailable, gatekeeper	Disclose nothing, get callback time, end
P4 Soft objection	Busy, not now, not interested	One reschedule offer, then close
P5 Normal flow	Identity, reminder, outcome, close	LLM phrases the reply within the current stage

A. Connecting the call

Case	Handling
Voicemail or answering machine	Detect via Twilio answering-machine detection or transcript (patterns exist at <code>main.py:149</code>). Leave a limited-content message or hang up; count as an attempt

Case	Handling
IVR or phone tree	Detect "press 1" prompts, hang up, record IVR_REACHED
Call screening ("what are you calling about")	Give name and organization only
Silence after answer	"Hello, can you hear me?" at 3 s, again at 8 s, end at 15 s
Busy, no answer or failed	Status callback, then retry policy (section H)
Call drops mid-conversation	Record DROPPED; one later retry, counted toward the cap

B. Identity and third parties

Case	Handling
Right person confirmed	Continue to the reminder
Someone else answers	Disclose nothing about the account; ask for a good callback time
Account holder unavailable	Offer callback; save the time as do-not-call-before
Wrong number	Apologize, mark the number invalid across campaigns, end
Deceased	Condolence line, stop all calls, escalate to back office
Refuses verification	Offer an official callback number; don't push
"Who is this?" before verification	Organization name and callback number only

C. The reminder itself

Case	Handling
Already paid	Thank them, optionally ask the date, flag for reconciliation, close. Never argue
Disputes the amount	Stop the reminder, record the dispute, escalate (P2)
Can't pay or hardship	Empathy, no pressure, escalate or offer callback
Wants a payment plan	Escalate; the bot doesn't negotiate
Wants to pay now	Never take card details on a call. Send a secure link or transfer
Asks for balance or date not on file	Say it isn't available; never invent; offer callback
Wants details in writing	Confirm email, text or mail (reuses the email flow)
Reschedule or change service	Get a specific date and time, confirm once, close

D. Consent and opt-out

Case	Handling
"Stop calling" or any reasonable wording	Fixed confirmation, add to internal do-not-call list (FCC: honor within 10 business days), end
Partial opt-out ("not at work", "text only", "after 5pm")	Save the preference, confirm, continue or close. Not a refusal
"How did you get my number?"	"It's the number on file with the organization"; offer to update or remove
Attorney or bankruptcy mentioned	P0: stop, take attorney contact if offered, escalate

E. Trust and AI

Case	Handling
"Are you a robot?"	Short honest disclosure, then continue or offer a human

Case	Handling
Suspects a scam	Direct them to the number on their bill; never ask for sensitive info
"Ignore your instructions"	Ignore it; repeat the current-stage question

F. Conversation mechanics

Case	Handling
"Hold on" or "one sec"	Hold mode: silence doesn't count as inactivity; "Take your time" once; check back at 30 s; end at 90 s
Talking to someone else in the room	Ignore low-confidence speech during hold
"What?" or "Say that again"	Replay the last reply word for word
"Slow down"	Lower TTS speed for the rest of the call
Not understood twice	Third time: offer callback or a human
Asks for another language	Record it, transfer or schedule a callback in that language, end
Chit-chat or off-topic	One short friendly reply, back to the current stage
Profanity or anger	De-escalate once; second time goes to P1 or a polite end
Says "bye" mid-call	End gracefully. Today it counts as a refusal (<code>intent.py:356</code>)
Long monologue	Summarize and ask one question

G. Human escalation and queueing

- **Agent available:** "Let me connect you," then a warm transfer with a short summary (Twilio `<Dial>` or a conference).
- **No agent free:** offer hold (about 60 s maximum, reassurance at 30 s) or a callback with a time window and confirmed number. Create a callback ticket.
- **After hours:** callback ticket for the next business day.
- **Transfer fails:** apologize, create a callback ticket, end. Never leave dead air.

H. Spam prevention

- **Before every dial:** consent on file; not on the national or internal do-not-call list; recipient's local time between 8am and 9pm; respect any do-not-call-before time; frequency cap (7 attempts per 7 days per account if Regulation F applies, stricter for your own campaigns).
- **Retry backoff:** no answer, +4 h; voicemail, +24 h; maximum attempts per campaign.
- **Deduplication:** remove duplicate leads across campaigns; one active call per number.
- **Caller ID:** consistent number with STIR/SHAKEN and branded caller ID to avoid "Spam Likely" labels.
- **In-call:** never ask the same question more than twice, cap total re-asks, keep the call-length limit, one reschedule offer per objection.

I. System failures

Case	Handling
LLM, TTS or STT error or timeout	Cached "Sorry, we're having trouble, we'll call you back," callback ticket, end
Reply slower than 1.5 s	Filler (existing); same fallback after 5 s

Part 3: Prompt and flow restructure

Code should make every call decision and the LLM should only phrase replies, which removes the redundancy below.

Redundancies found

Duplication	Where
Follow-up questions come from three sources	Prompt MANDATORY FOLLOW-UP, special_instruction, and VOB questions appended after generation (llm.py:185)
Email confirmation implemented three times	main.py:1127, :1178, :1343
Stage inferred from turn count, not events	Turn 3 marks the pitch delivered (main.py:1527)
Voice rules block appears twice	language.py:1257 and :1298
Overlapping instructions	Name rules in two places; tone vs delivery style; leftover sales "social proof" and "urgency" notes
Old use case still wired in	VOB follow-ups, billing-audit intents, call_type: VOB, meeting/BAA outcomes in logger.py

Target design

The policy engine is the only place that decides what happens next; the LLM never decides flow.

Transcript

```
-> Classifier (regex first, then small LLM)
-> Policy engine (priority ladder + call state)
    |- SAY_TEMPLATE -> cached audio
    |- ASK_SLOT / RESPOND -> LLM phrases reply -> TTS
    |- ESCALATE -> transfer or callback
    '- END -> disposition logged
```

- Classifier with reminder intents:** CONFIRM_IDENTITY, DENY_IDENTITY, THIRD_PARTY, UNAVAILABLE, ALREADY_PAID, DISPUTE, HARDSHIP, PAYMENT_PLAN, PAY_NOW, RESCHEDULE, CALLBACK_LATER, WRITTEN_REQUEST, HOLD, REPEAT, SLOW_DOWN, HUMAN_REQUEST, DNC, PARTIAL_OPT_OUT, WRONG_NUMBER, ATTORNEY, BANKRUPTCY, DECEASED, ABUSE, DISTRESS, ASK_IF_AI, TRUST_CONCERN, LANGUAGE_REQUEST, VOICEMAIL, IVR, SMALLTALK, GOODBYE, UNCLEAR. Keep regex first and the small LLM second.
- Policy engine (new module):** takes intent plus call state and returns one action: SAY_TEMPLATE(id), ASK_SLOT(slot), RESPOND(goal), ESCALATE(kind) OR END(disposition).
- Explicit call state:** slots identity_verified, reminder_delivered, outcome (paid, rescheduled, callback, dispute, escalated, opted_out, wrong_number) and preferences; counters reask, clarify, objection, ai_ask, hold_active. The stage follows from the slots: INTRO, VERIFY, REMIND, RESOLVE, CLOSE.
- One contact-confirmation handler** replaces the three email copies.
- One final disposition per call** from the policy engine, replacing SUCCESS_MEETING and BAA outcomes in logger.py.

Prompt skeleton

Fixed content first (cache-friendly), short dynamic content last:

```
[FIXED]
Persona and voice rules (once)
Compliance: never invent amounts, never take card data,
  honest AI disclosure, no third-party disclosure
Output format rules
[DYNAMIC]
STAGE: REMIND GOAL: deliver reminder, ask if already paid
FACTS: name, organization, due date/amount (only if on file)
STATE: identity_verified=yes; filler_played=yes; avoid openers: [...]
INTENT: ALREADY_PAID -> guidance (1-2 lines)
LIMIT: 35 words, exactly one question unless CLOSE
```

All intent guidance lives in one table. Delete the post-generation follow-up append in `llm.py`, the duplicate voice block, and the sales notes.

Rollout and decisions

Each step ships behind a flag, so current behaviour holds until a step's numbers and tests pass.

Rollout order

- 1 **Measure and baseline.** Add Step 0 latency timestamps. Build a text-level pytest harness that feeds scripted transcripts through classifier, policy and prompt with no audio. Write one test per catalog row and record today's behaviour as the baseline.
- 2 **Latency Step 1** safe wins.
- 3 **Policy engine in shadow mode.** Log its decision next to the current logic on real calls and compare.
- 4 **Switch over.** New intents, P0/P1 fixed lines, slimmed prompt; delete the redundant code paths.
- 5 **Latency Steps 2 and 3** behind flags, compared on p50 and p95.
- 6 **Dialing-side rules.** Spam prevention (H) and the escalation queue (G); these need a lead and call database and a transfer target.

Decisions needed

- Are reminders for money owed on behalf of another organization? This decides whether FDCPA and Regulation F apply.
- Is there a real human transfer destination (number or queue), or does escalation mean callback only?
- Is SMS or email available for payment links and written follow-ups?
- Approve honest AI disclosure in place of the current denial instruction.

Sources

Findings from web search result summaries; confirm legal points with counsel.

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<https://futureagi.com/blog/how-to-optimize-voice-agent-latency-2026/>